

Design CampusEats — Services, Contracts & Schema

Goal

Produce the **design benchmark** for CampusEats — its services, the contract each one exposes, and the database schema behind them.

Approach

For CampusEats identify **capabilities** → **group by data ownership** → **draw boundaries** → **define contracts**. Then extend the same boundaries into a database schema. Use **draw.io (diagrams.net)** for both diagrams so they are editable and can serve as the benchmark.

Tasks

- From your Assignment 1 brief, list every distinct job the system does — at least six Capabilities (the verbs).
Submit — *the capability list in design.pdf*.
- Service design diagram (draw.io)**. Draw your 3–6 services as boxes; label each box with the data it **owns**. Add an arrow for every cross-service call, labelled with the operation used (e.g. Orders → Catalogue: `checkItem`). No two services may share data. This diagram is the benchmark you will build to match. Mention team member along with each identified Service.
Submit — *services.drawio (editable source) and services.png/pdf (exported)*.
- For each service, write a **contract**: a table of the operations other services may call — operation name, input, output, errors. Do not mention tables, id formats, or SQL; expose the operation, hide the inside.
Submit — *the contract tables in design.pdf (≥3 operations for your two main services, ≥1 for the rest)*.
- Specify your central operation — `placeOrder` — in full: its inputs, outputs, error cases, and the internal details it must hide from callers.
Submit — *the full specification in design.pdf*.
- Database schema (draw.io ER diagram)**. For each service, design the tables that hold the data it owns: table name, columns with types, primary key, and foreign keys. Keep the boundary rule — **no table may belong to two services**. Draw it as an entity-relationship diagram, and also write the equivalent `CREATE TABLE` statements.
Submit — *schema.drawio and schema.png/pdf (the ER diagram), plus schema.sql (the CREATE TABLE sketch)*.
- Check each service against the five properties of a Service (reachable, self-contained, has a contract, independent, loosely coupled). Where one fails, say so and note how you would fix it.
Submit — *the validation table in design.pdf*.

What to submit

Commit all of the following into your **campuseats** repository, and submit the link:

- design.pdf — services, members responsible for implementing the services, capabilities, contracts, the full operation, and the validation table (Tasks 1, 3, 4, 6).
- services.drawio + services.png — the service design benchmark (Task 2).
- schema.drawio + schema.png + schema.sql — the database schema (Task 5).